

Student Expectations Document

Advanced Apex Project-2 (M.Sc. DS & AI)

Cohort-1 | Trimester-3

1. Purpose of the Project

The Advanced Apex Project-2 is designed to help students demonstrate their understanding of the complete Machine Learning workflow through meaningful experimentation, analysis, and interpretation.

Students are expected to focus on:

- Correct implementation of concepts
- Structured experimentation
- Understanding model behaviour
- Interpreting results
- Maintaining reproducibility and readability

The project is experimentation-oriented and not deployment-oriented.

2. General Expectations from Students

Students are expected to:

- Follow a clear end-to-end workflow
- Maintain a single consolidated notebook
- Include markdown explanations between code cells
- Explain preprocessing and modelling decisions
- Perform meaningful experimentation
- Compare models appropriately
- Interpret evaluation metrics and results
- Submit clean, organized, and reproducible work

Students should demonstrate:

- Sincere implementation effort
- Logical workflow understanding
- Progressive model improvement
- Proper experimentation practices

3. What Students Are Expected to Cover

A. Problem Understanding

- Clearly explain the selected problem statement
- Define the business objective
- Explain the expected outcome of the project

B. Dataset Understanding

- Describe the dataset source

- Explain dataset size and features
- Identify the target variable (if applicable)
- Perform initial dataset inspection
- Mention expected challenges or limitations

C. Data Preparation & Preprocessing

- Missing value handling
- Duplicate removal
- Datatype correction
- Encoding
- Scaling/normalization (if required)
- Outlier handling (if necessary)

D. Exploratory Data Analysis (EDA)

- Create meaningful visualizations
- Identify trends and patterns
- Analyse correlations and distributions
- Provide observations and interpretations

E. Feature Engineering

- Create meaningful new features
- Transform existing features
- Select important features
- Justify if FE is not required

F. Model Building

- Implement at least one baseline model
- Compare with improved models
- Use appropriate algorithms

G. Hyperparameter Tuning

- Perform beginner-to-intermediate tuning
- Use GridSearchCV, RandomizedSearchCV, or manual tuning
- Explain tuning observations

H. Model Evaluation

- Use appropriate metrics
- Interpret model performance
- Compare models meaningfully

I. Interpretation & Insights

- Explain model behaviour
- Analyse improvements
- Discuss overfitting/generalization

J. Documentation & Reproducibility

- Properly organized notebook
- Runnable code
- Clear markdown explanations

4. Submission Expectations by Stage

Week-2: Project Outline Submission

Deliverables:

- Project Outline Report (PDF)
- Initial Notebook (.ipynb)

Expected Coverage:

- Problem statement
- Dataset details (Data dictionary)
- Proposed workflow
- Baseline model
- Evaluation metrics
- Tuning strategy

Week-5: Progress Submission

Deliverables:

- Progress Report (PDF)
- Working Notebook (.ipynb)

Expected Coverage:

- Preprocessing progress
- EDA
- Initial feature engineering
- Baseline modelling
- Initial comparison
- Preliminary tuning and evaluation

Week-9: Final Submission

Deliverables:

- Final Project Report (PDF)
- Final Consolidated Notebook (.ipynb)

Expected Coverage:

- Complete workflow
- Multiple model experimentation
- Hyperparameter tuning
- Model comparison
- Evaluation and interpretation
- Final conclusions

5. ANN Project Expectations

For ANN-based projects, students should:

- Briefly explain ANN architecture
- Analyse training behaviour
- Compare training vs validation performance
- Observe convergence trends
- Discuss overfitting lightly

Deep theoretical discussion is not required.

6. Recommended Notebook Structure

1. Problem Statement & Objective
2. Dataset Description
3. Dataset Loading
4. Data Inspection
5. Data Preparation & Feature Engineering
6. Exploratory Data Analysis
7. Baseline Model Construction
8. Improved Model(s)
9. Hyperparameter Tuning
10. Evaluation Metrics
11. Model Comparison
12. Interpretation & Discussion
13. Overfitting / Generalization Discussion
14. Conclusion
15. References

7. Evaluation Philosophy

Evaluation will focus on:

- Workflow correctness
- Understanding of concepts
- Experimentation effort
- Model improvement attempts
- Interpretation quality
- Documentation and reproducibility

Students are NOT expected to:

- Achieve state-of-the-art performance
- Build dashboards
- Deploy models
- Use advanced research-level optimization
- Implement MLOps pipelines

Meaningful experimentation and sincere effort are valued more than complexity.

8. Final Note to Students

Students are encouraged to:

- Experiment confidently
- Learn from model behaviour
- Document observations clearly
- Focus on understanding rather than perfection

A well-structured and sincerely executed project with proper explanations and experimentation will be valued positively during evaluation.